Task NO: 1a

Task No: Running Python Script and various expressions in an interactive interpreter
key terms converod: Introduction to python, commands, script
Date: 06/08/25

Aim: To write a python program that calculate the total amount spent by karan on books, groceries and transport

Algorithm:

1. start the program.
2. Accept the amount spent on books, groceries, and transport.
3. Calculate the total express by summing all three amounts.
4. Display that total amount spent.
5. end the program.

Result:

The program was successfully excuted and the total amount spent by karan was calculated and displayed as expected.

Program:-

```
# program to calculate that expenses of karan.
# step 1: Assign expenses
    books = 150
    groceries = 220
    transport = 90
# step 2: calculated total.
  total - expenses = books + groceries + transport.
# step 3: Display the result
 Print f(" Total expenses incurred by karan: ₹", total expenses)
```

Input:

Books = ₹150

Groceries = ₹ 220

transport = ₹ 90

output:

total expenses incurred by karan: ₹460

Task No:- 1b

Date : 06/08/25

Aim:- To write a python program that calculates and displays the Body Mass Index (BMI) of a person using their weight (in kilograms) and height (in meters).

Algorithm:

1. Start the program.
2. Prompt the user to input their weight in ~~kilometers~~ kilograms
3. Prompt the user to input their height in meters.
4. Calculate the BMI using the formula:

$$BMI = \frac{weight}{height^2}$$

5. Display the calculated BMI
6. End the program.

Result:-

The program was successfully executed and the total body mass Index of a person was calculated and displayed as expected.

Program:

```
#BMI calculator

# step 1: Get input from the user.

# weight = float (input ("Enter your weight in kilograms:"))

heigh = float (input("enter your height in meters:"))

# Step 2: calculate BMI

bmi = weight /(height**2)

#step3 : Display result

Print ("your body mass Index (BMI) is :", round (bmi, 2))
```

Input:

Enter your weight in kilograms: 70

Enter your heigh in meters : 1.75

output:

your body mass Index (BMI) is: 22.86

Task No:-1C

Date: 06/08/25

Aim:- To write python program to find the area of triangle when the lengths of all three sides are given, using Heron's formula.

Algorithm:

1. Start the program.
2. Accept or assign the length of the three sides: a, b and c.
3. Calculated the semi-perimeter

$$s = \frac{a+b+c}{2}$$

4. use heron's formula to calculat the area

$$Area = \sqrt{s(s-a)(s-b)(s-c)}$$

5. Display the area of the triangle
6. End the program.

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Result:- The program was successfully executed and the area of the triangle using heron's formula was calculated and displayed as expected.

Program:

```
import math

# step 1: Assign side lengths.

a=8
b=6
c=4

# step 2: calculate semi-perimeter.

    s=(a+b+c)/2

# step 3: Apply Heron's formula.
  area = math.sqrt(s*(s-a)*(s-b)*(s-c))

# step 4: Display result

print("The area of the triangle is :", round(area, 2), "square cm")
```

Input:

side a = 8cm

side b = 6cm

side c = 4cm

output:-

The area of the triangle is: 11.62 sq units